ness of the machine or the complexity of writing an algorithm for it.

Instead of presenting general concepts of computer system design, the author nicely gives some "inside" feelings of these concepts by means of concrete examples taken from the HP 2116 and System 360. Thus the text supplemented by reference manuals of these computers should provide the student with an excellent understanding of computer organization, assembly language programming, and data structures.

The author covers a lot of concepts including paging, recursive programs, and input-output programming which most authors generally avoid. While the explanations are clear, much of the detail has been sacrificed for conciseness. Thus it may be difficult for the reader to evaluate different design opitions for a particular goal. Carefully chosen examples supplement the explanations. The chapter on input-output operations is simply excellent.

While it is very hard to venture any suggestions of improvement on such an excellent text, the writer would welcome the following topics in a future edition: the memory system; control units; microprogramming; arithmetic and logical processing; etc.

In summary, Prof. Stone must be congratulated for providing an excellent self-contained introductory text on computer organization and data stuctures.

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B73-10 Time Sharing Systems—G. M. Bull and S. F. G. Packham. (New York: McGraw-Hill, 1972, 140 pp., $9.95.)

The book is a general and fairly brief introduction to the structure of time-sharing computer systems and to the facilities that they provide. The authors indicate that their book is supposed to serve two purposes. It is supposed to assist in making comparisons between different systems in order to make a selection evaluation, and it should also serve as a textbook for an undergraduate course on time-sharing systems.

The text is divided into six chapters. In an introductory chapter, time-sharing computer systems are defined and some of their characteristics and advantages are outlined. A very brief introduction to operating systems and to the historical development of time-sharing completes the first chapter. The second chapter presents the user facilities that are commonly available or desirable in time-sharing systems. User command and programming languages, filing systems, file editors, debugging aids, and terminal devices are reviewed with examples drawn mostly from the time-sharing system at Dartmouth College. Chapter 3 is a brief introduction to the operating systems techniques that have been used to support time-sharing systems, including roll-in/roll-out systems, systems with a fixed-memory partition for each job, and paging; a description of the communication interface between the computer and the terminals is also given. Chapter 4 begins with a description of multiprogramming and its implementation via hardware interrupts and supervisor calls; this is followed by a discussion of the use of scheduling techniques to implement multiprogramming. A section is devoted to a review of virtual memory techniques and to some of the problems that arise in this area, such as the choice of page size and the page replacement problem. Some sections of this chapter are devoted to problems of system architecture and to the communications processors in a time-sharing system. Chapter 5 is devoted to a brief discussion of resource allocation, scheduling, and system failures. In the last chapter, the authors outline the main steps that they recommend in a selection evaluation of a time-sharing system and they present a case study.

In the reviewer's opinion, the book presents a very superficial view of time-sharing systems and of their evaluation; the contents of the first two chapters are obvious to a casual user of a time-sharing system and the remaining parts are of low technical content. Some sections in which technical problems are discussed, such as the choice of page size and of page replacement algorithms, do not seem to take into consideration the more recent studies on the subject. Similarly, the last chapter of the book which is devoted to selection evaluation contains some general advice and specifications without going to the more technical and interesting matters involved in such a study.

The book cannot be recommended as a textbook at the graduate or undergraduate level in a university curriculum. It must also be added that it compares rather badly with the books *Time Sharing Computer Systems* by Wilkes and *Time Sharing Design Concepts* by Watson. Finally, we would like to mention that the bibliography given at the end of the book omits some of the earlier classical papers on multiprogramming and time-sharing systems, as well as some of the more recent papers on performance evaluation.

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B73-11 Digital Circuits and Devices—T. Kohonen. (Englewood Cliffs, N.J.: Prentice-Hall, 1972, 462 pp., $16.50.)

This book comprises two distinct parts. The first five chapters form a standard exposition on logic design, while the remainder, which is of about the same length as the first part, deals with digital electronics. Each chapter is followed by a set of problems and references, and a lengthy bibliography is included as an Appendix.

Chapter 1 introduces number representation, codes, and binary storage elements. Combinational circuits and switching algebra are taken up in Chapter 2. Simplification of single-output functions is treated by the Quine-McCluskey method and the Karnaugh map, and the chapter concludes with some examples of decoders and code converters. Chapter 3 is about sequential circuits, proceeding from basic types of flip-flops, through simple counters, to the standard design methods of asynchronous (flow table) and synchronous (excitation map) circuits. This reviewer would quibble with the treatment of master-slave flip-flops in that what is described (p. 112) as a "fast-gate" is more significantly a threshold-shifting gate that ensures master-slave isolation even when the clock signal has slow rise/fall times. Chapter 4 discusses various types of counter, digital arithmetic, including multiplication and division, and ends with a particularly good treatment of two devices not usually covered in such texts—the binary rate multiplier and digital differential analyzer. Finally, synchronous control design, especially as applied to a stored program processor, is the subject of Chapter 5.

Chapter 6, the first of the section on digital electronics, summarizes the basic ideas on current and voltage switches (which the student is presumed to have met before) and goes on to treat the salient features of bipolar and field-effect switching circuits. Common logic circuit families are discussed next in Chapter 7, which also contains a description of packaging techniques. At a time when silicon technology is developing so rapidly, the author is to be admired for his daring comparison of bipolar and unipolar devices. Chapter 8 is mostly devoted to various sorts of multivibrators, concluding with a short discussion of ramp circuits. Memories form the subject of Chapter 9. With the exception of core memories (2 D and 3 D), for which some electromagnetic features are presented, the discussion is on how the memories are organized. These include semiconductor (static) RAM and ROM and magnetic surface memories. The book concludes with a chapter on transmission lines, with examples of line drivers and receivers.

The material is presented in a logical and orderly fashion and the author has struck a nice balance, on the whole, between theory and practice. But the book is not without its peculiarities. This reviewer would question the wisdom of treating on an equal footing both logic